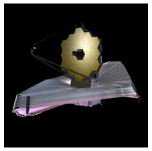

5564 - Measuring the Hubble constant with the next multiple-imaged lensed supernova

Cycle: 3, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
imaging				
	1	Imaging AT2025wny	NIRCam Imaging	(2) AT2025wny
spectroscopy				
	2	IFU AT2025wny	NIRSpec IFU Spectroscopy	(2) AT2025wny

ABSTRACT

Spectroscopic time-delay measurements of multiply-imaged supernovae offer a very efficient way to measure the expansion rate of the universe. NIRCam and NIRSpec follow-up observations with JWST upon the next lensed supernova (SN) discovered by the Zwicky Transient Facility (ZTF) or the La Silla Schmidt Southern Survey (LS4) could pave the way for a novel technique to measure the Hubble constant and adress the "Hubble tension". Unlike any other time-delay measurement, spectroscopic dating of multiple SN images can be done without repeated observations. We propose to select a suitable lensed supernova for ToO observations and anticipate a 5% (or better) measurement of H0, with only 3.5 hours of JWST time.

OBSERVING DESCRIPTION

We propose to carry out non-disruptive ToO observations of a strongly lensed, multiply-imaged, supernova discovered using wide-field time-domain surveys. Thanks to the unique combination of near-IR sensitivity, througput and angular resolution, JWST observations will allow us to measure positions and spectroscopic time-delays between resolved multiple images, the latter with an accuracy of about one day. We will accomplish this through SED dating of the individual resolved SN images, with the aim to establish a novel and powerful technique to measure the Hubble constant with strongly lensed supernovae. The requested observations will allow us to measure H0 with 5% accuracy or better. This is particularly timely given the "Hubble tension", arguably the hottest issue in cosmology today. The JWST observations are to be complemented with two visits of (mainly) optical imaging with HST, separated by two to four weeks. This will allow us to test the robustness of spectroscoic time-delay measurements and account for extinction affecting the image flux-ratios and the inferred total magnification.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	AT2025wny	RA: 07 16 34.5053 (109.1437721d) Dec: +38 21 6.97 (38.35194d) Equinox: J2000	Epoch of Position: 2000	

Comments: Transient in the ZTF survey in a known galaxy-scale lensing candidate system (lens at z=0.3754) . Liverpool telescope images from Oct 3 and 4 revelaled two additional fainter images, indicating the lensing nature of the transient.
Category=Star
Description=[Supernovae]
Extended=NO

Observation

Proposal 5564, Observation 1: Imaging AT2025wny Wed Oct 22 16:00:10 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	AT2025wny	RA: 07 16 34.5053 (109.1437721d) Dec: +38 21 6.97 (38.35194d) Equinox: J2000	Epoch of Position: 2000	

Comments: Transient in the ZTF survey in a known galaxy-scale lensing candidate system (lens at z=0.3754) . Liverpool telescope images from Oct 3 and 4 revelaled two additional fainter images, indicating the lensing nature of the transient.
Category=Star
Description=[Supernovae]
Extended=NO

Template

Module	Subarray	Target Placement
ALL	FULL	Module Gap

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	INTRAMODULEBOX	2	SMALL-GRID-DITHER		2

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F115W	F277W	BRIGHT1	3	2	8	4	472.418	
2	F150W	F277W	BRIGHT1	3	2	8	4	472.418	

Special Requirements

Offset 121.0 arcsec, 35.0 arcsec
Target Of Opportunity Response Time 21 Days, Carry-Over

Observation

Proposal 5564, Observation 2: IFU AT2025wny Wed Oct 22 16:00:10 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	AT2025wny	RA: 07 16 34.5053 (109.1437721d) Dec: +38 21 6.97 (38.35194d) Equinox: J2000	Epoch of Position: 2000	

Comments: Transient in the ZTF survey in a known galaxy-scale lensing candidate system (lens at z=0.3754) . Liverpool telescope images from Oct 3 and 4 revelaled two additional fainter images, indicating the lensing nature of the transient.
Category=Star
Description=[Supernovae]
Extended=NO

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G140M/F100LP	NRSIRS2RAPID	85	1	false	true	NONE	4	4	5018.578	27082

Special Requirements

Target Of Opportunity Response Time 21 Days, Carry-Over